Tungsten TotalAgility
Integration with ZohoWorkDrive

Pocket Guide

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TUNGSTEN AUTOMATION

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Version History

02/17/2026	Version 1	Initial version. Documented all the steps to get access to ZohoWorkdrive
02/24/2026	Version 2	Updated all steps to upload Documents from TotalAgility to Zoho WorkDrive.
03/06/2026	Version 3	Final version. Updated Zoho account set up and document upload along with Metadata to ZohoWorkdrive from Totalagility

Pocket Guide: Integrating ZohoUploader.dll with TotalAgility Cloud for Zoho WorkDrive Uploads along with steps to set up Zoho Account.

Overview

This guide provides step-by-step instructions for deploying and configuring the ZohoUploader.dll in TotalAgility Cloud, enabling seamless document uploads from KTA folders to Zoho WorkDrive with robust error handling, security, and best practices.

The ZohoUploader.dll enables integration between TotalAgility Cloud (KTA) and Zoho WorkDrive. It exposes a single public method, UploadFolderDocuments, which uploads documents from a specified KTA folder to a Zoho WorkDrive folder, generating JSON metadata for each document. This integration is designed for use within a TotalAgility Script Task and supports robust logging and error handling.

Prerequisites & Dependencies

Requirement	Details
TotalAgility Cloud Access	Admin access to TTA Cloud Designer
Zoho WorkDrive Account	API access with client credentials
ZohoUploader.dll	Compiled as a .NET Class Library with all dependencies included
Newtonsoft.Json	Must be referenced in the DLL project for JSON serialization
API Credentials	Zoho Client ID, Client Secret, and Scope

Note: All dependencies (including Newtonsoft.Json) must be bundled with the DLL at build time, as cloud environments do not support runtime NuGet installation.

Zoho WorkDrive Setup Guide

Complete step-by-step instructions to create a Zoho account, access WorkDrive, manage folders, and configure API credentials.

Step 1: Create Zoho Account

1. Visit the Zoho sign-up page at https://www.zoho.com/mail or https://accounts.zoho.in
2. Click on "Sign Up" or "Create Account"
3. Fill in the required details:
   - First Name and Last Name
   - Desired email username (e.g., username@zohomail.com)
   - Create a strong password
   - Enter your Indian mobile number in +91 format
4. Agree to the Terms of Service and Privacy Policy
5. Click "Sign Up" or "Create Account"
6. Enter the OTP (One-Time Password) sent to your mobile number
7. Complete additional setup:
   - Select your preferred language
   - Choose timezone (Asia/Kolkata for India)
   - Configure any additional preferences
8. Your Zoho account is now created and ready to use

Step 2: Access Zoho WorkDrive

1. Log in to your Zoho account at https://accounts.zoho.com
2. Navigate to WorkDrive using one of these methods:
   - Visit directly: https://workdrive.zoho.com
   - Or search for "WorkDrive" in your Zoho applications dashboard
   - Or access from Zoho One menu (if subscribed)
3. WorkDrive interface will load showing your folders and team spaces
4. You'll see "My Folders" section for personal files
5. Team folders will appear if you're added to any teams

Step 3: Create a Folder in WorkDrive

1. In the WorkDrive interface, navigate to "My Folders" or select a team folder location
2. Click on the "New Folder" button (usually a "+" icon or "New" button)
3. Enter a descriptive name for your folder (e.g., "Project Documents" or "API Integration Files")
4. Click "Create" or press Enter
5. Your new folder will appear in the selected location
6. You can now organize files and subfolders within this folder

Step 4: Copy the Folder ID

1. Open the folder you just created by clicking on it
2. Look at the URL in your browser's address bar
3. The URL will look like: https://workdrive.zoho.com/.../folders/{folder_id}
4. Copy the long alphanumeric string after "/folders/"
    - This is your Folder ID (typically 40+ characters long)
    - Example: a1b2c3d4e5f6g7h8i9j0k112m3n4o5p6q7r8s9t0
5. Save this Folder ID securely:
    - Create a text file or note document
    - Label it clearly as "Zoho WorkDrive Folder ID"
    - Store in a secure location (password manager, encrypted note, etc.)

Alternative Method: Right-click on the folder name in the sidebar,select "Properties" or "Info," and the Folder ID may be displayed in the details panel.

Step 5: Access Zoho API Console

1. Open a new browser tab and visit the Zoho API Console:
    - For US Data Center: https://api-console.zoho.com
    - For EU Data Center: https://api-console.zoho.eu
    - For India Data Center: https://api-console.zoho.in

2. Log in using your Zoho account credentials
3. You'll be directed to the API Console dashboard

4. This is where you'll manage API clients and credentials

Note: Make sure to use the correct data center URL based on where your Zoho account is registered (typically India DC for Indian users).

Step 6: Create Self Client (Self Service Application)

1. In the API Console, click on "Get Started" (if first time) or "Add Client" button
2. You'll see different client type options
3. Hover over or click on "Self Client" option
4. Click "Create Now" or "Create Self Client"
5. A confirmation dialog will appear
6. Click "Create" to confirm
7. Click "OK" on the success message
8. Your self client is now created

What is Self-Client? A self-client allows your application to access Zoho services on behalf of itself (not end-users), ideal for server-to-server integrations.

Step 7: Copy Client ID and Client Secret

1. After creating the self-client, you'll be on the client details page
2. You'll see two tabs or sections:
   - "Client Secret" tab (default view)
   - Shows Client ID and Client Secret
3. Copy the Client ID:
   - Click the "Copy" icon next to Client ID
   - It's a long alphanumeric string (e.g., 1000.ABC123XYZ456...)
4. Copy the Client Secret:
   - Click the "Copy" icon next to Client Secret
   - This is also a long alphanumeric string
   - **IMPORTANT:** This is shown only once initially - copy it immediately
5. Save both credentials securely:
   - Create a secure text file or use a password manager

- Label clearly: "Zoho API Client ID" and "Zoho API Client Secret"
- Store in an encrypted location
- Never share these credentials publicly or commit to version control

Step 8: Configure API Scopes for WorkDrive (Optional)

If you need to configure specific scopes for WorkDrive API access:

1. Common WorkDrive API scopes include:
    - WorkDrive.files.ALL - Full access to files
    - WorkDrive.files.READ - Read-only access to files
    - WorkDrive.files.CREATE - Create files
    - WorkDrive.files.UPDATE - Update files
    - WorkDrive.files.DELETE - Delete files
    - WorkDrive.team.READ - Read team information
    - WorkDrive.teamfolders.READ - Read team folders
    - WorkDrive.workspace.ALL - Full workspace access
    - WorkDrive.organization.READ - Read organization details
2. These scopes are used when generating access tokens
3. You'll specify required scopes in your API authentication code
4. For most integrations, WorkDrive.files.ALL provides comprehensive access

Summary Checklist

- Zoho account created and verified
- WorkDrive accessed successfully
- Folder created in WorkDrive
- Folder ID copied and saved securely
- API Console accessed
- Self Client created
- Client ID copied and saved securely
- Client Secret copied and saved securely
- All credentials stored in secure location
- Ready to integrate WorkDrive API

Security Best Practices

- Store all credentials (Folder ID, Client ID, Client Secret) in encrypted storage
- Use environmental variables or secure vault systems for production applications
- Never commit credentials to version control systems (Git, SVN, etc.)
- Restrict API scope permissions to minimum required for your use case
- Regenerate Client Secret periodically for enhanced security
- Monitor API usage through the API Console dashboard
- Implement proper error handling and logging in your integration code

Next Steps

After completing this setup:

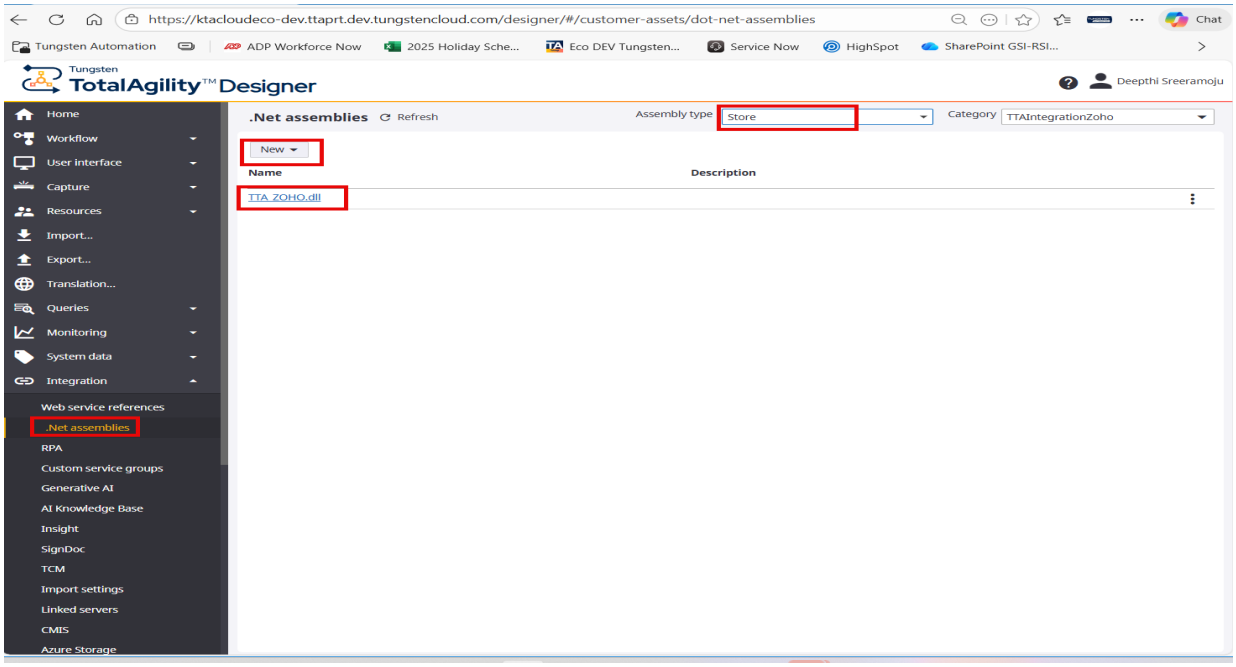
1. Implement OAuth 2.0 authentication flow in your application
2. Generate access tokens using Client ID and Client Secret
3. Make API calls to WorkDrive endpoints using the Folder ID
4. Test file upload, download, and management operations
5. Implement error handling and retry logic
6. Monitor API quota and usage limits

Support Resources- Zoho

- Zoho WorkDrive API Documentation: https://workdrive.zoho.com/apidocs/v1/
- Zoho API Console: https://api-console.zoho.in
- Zoho Developer Community: https://help.zoho.com/portal/en/community/
- Zoho WorkDrive Support: support@zohoworkdrive.com

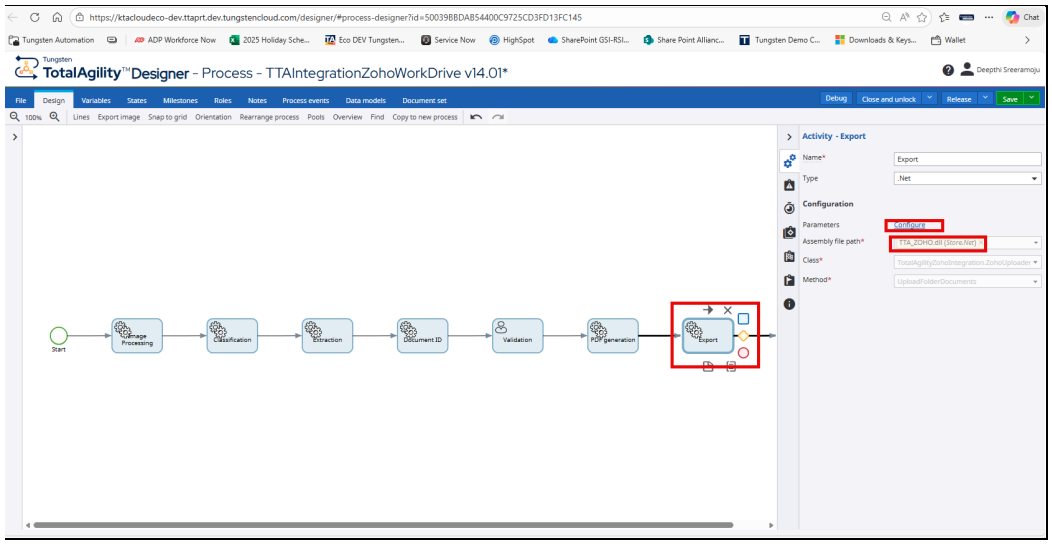
Deploying ZohoUploader.dll in TotalAgility Cloud

1. Build the DLL
    - Ensure ZohoUploader.dll is built as a Class Library.
    - Reference Newtonsoft.Json in your project.
    - Bundle all dependencies in the output directory.
2. Register the DLL in TotalAgility
    - Navigate to Integration:
    - Go to Integration > .Net Assemblies in the TotalAgility Designer.
    - Add the DLL:
    - Click the + button and upload ZohoUploader.dll.
    - Confirm the addition; the DLL will now be available for script tasks.
3. Security Considerations
    - Only upload trusted, signed DLLs.
    - Do not hard-code sensitive credentials in the DLL.
    - Use secure storage for API secrets.

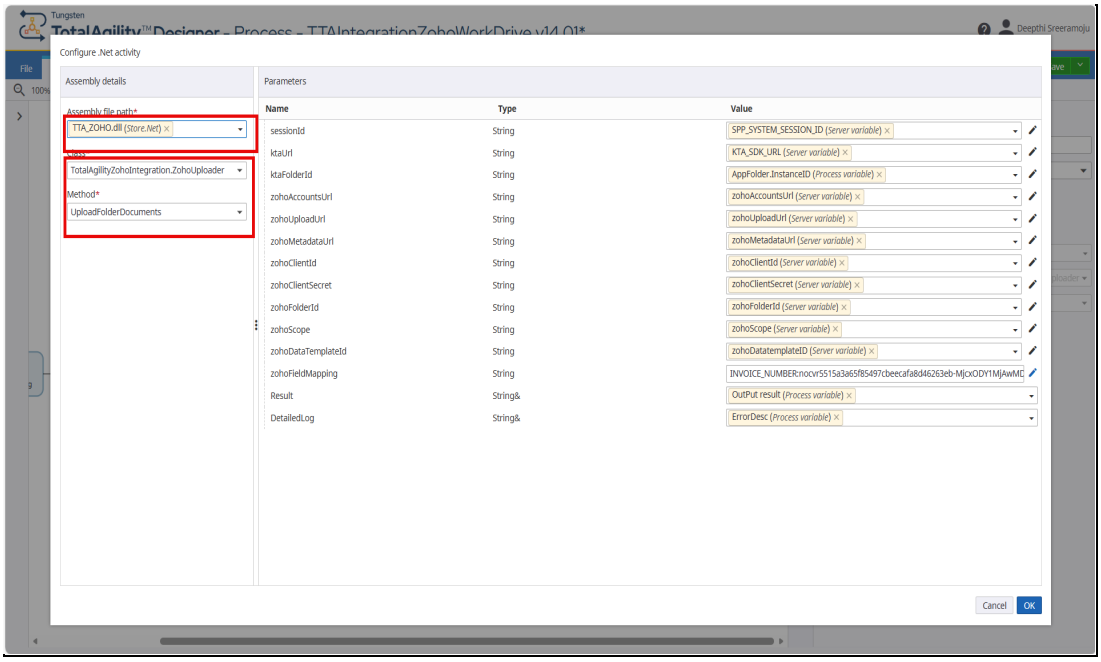


Configuring the Script Task

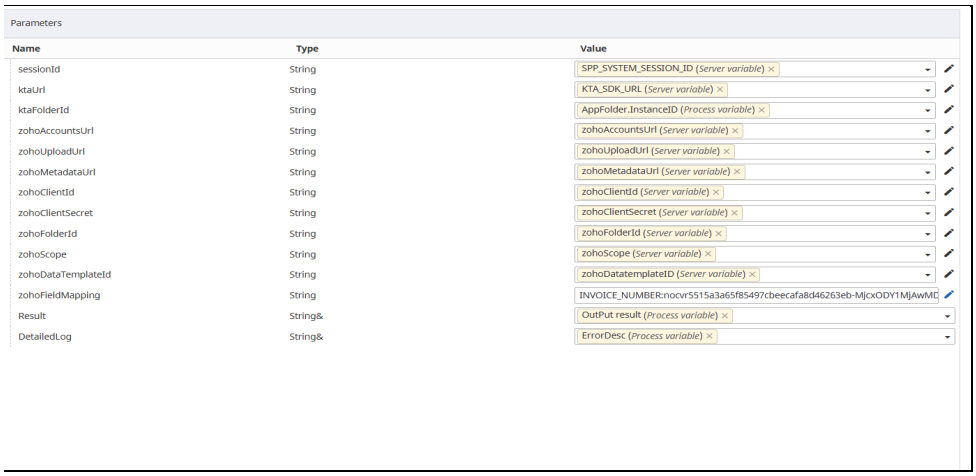
1. Add a .NET Script Activity to your process map.
2.



3. Configure the Script:
   - Class: ZohoUploader
   - Method: UploadFolderDocuments



4. Map Input/Output Parameters:
   - Link each method parameter to a process variable of the appropriate type.



Input & Output ParametersOutput Parameters

Input Parameters

Parameter	Description	Example Value
sessionId	KTA session ID	`SPP_SYSTEM_SESSION_ID`
ktaUrl	KTA server URL	`https://<tenant>.totalagility.com/TotalAgility/Agility.Server.Web`
ktaFolderId	ID of the KTA folder to upload from	`12345`
zohoAccountsUrl	Zoho Accounts URL	`https://accounts.zoho.com`
zohoUploadUrl	Zoho WorkDrive API upload URL	`https://workdrive.zoho.com/api/v1/upload`
zohoClientId	Zoho API Console Client ID	`your-client-id`
zohoClientSecret	Zoho API Console Client Secret	`your-client-secret`
zohoFolderId	Target folder ID in Zoho WorkDrive	`67890`
zohoScope	Zoho API scope	`WorkDrive.files.CREATE,WorkDrive.DataTemplates.CREATE
zohoMetadataUrl	Target customfields update values	
zohoDataTemplateId	Target Template to update the fields	`5214vdr67890`
zohoFieldMapping	Target Fields Values	InvoiceNumber:abc-001;InvoiceDate:abc-002;Total:abc-003

Output Parameters

Parameter	Description
Result	Summary of the upload process (e.g., "Upload complete. Success: X, Failed: Y")
DetailedLog	Detailed log of the upload process, including per-document status and errors

How It Works

1. Input Validation:
    - Ensures all required parameters are provided and valid.
2. Document Retrieval:
    - Fetches documents from the specified KTA folder.
3. Zoho Authentication:
    - Obtains an OAuth token using Zoho credentials and scope.
4. Document Upload:
    - Converts each document to PDF.
    - Uploads the PDF and its JSON metadata to Zoho WorkDrive.
5. Logging:
    - Logs the status (success/failure) of each document upload, including file size, Zoho file IDs, and errors.

Error Handling & Troubleshooting

- Result output will contain an error message if the process fails.
- DetailedLog provides per-document status and error details.

Common Errors:

- Invalid or expired Zoho OAuth tokens.
- Missing or incorrect input parameters.
- Network or API failures.

Tip: Always review the DetailedLog output for troubleshooting.

Example Usage

Input Example:

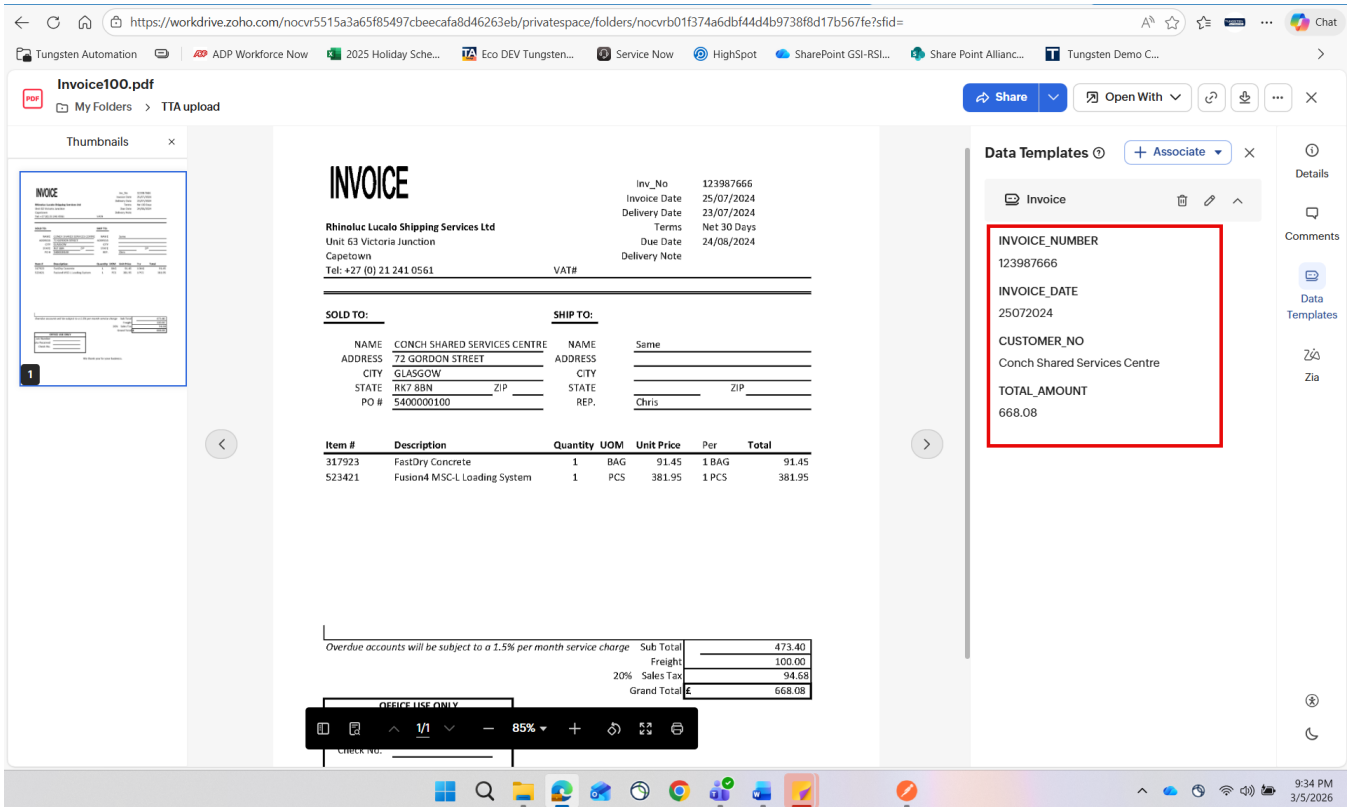
sessionId: "SPP_SYSTEM_SESSION_ID"
ktaUrl: "https://example.totalagility.com/TotalAgility/Agility.Server.Web"
ktaFolderId: "12345"
zohoAccountsUrl: https://accounts.zoho.com
zohoUploadUrl: https://workdrive.zoho.com/api/v1/upload
zohoMetadataUrl: https://www.zohoapis.com/workdrive/api/v1/custommetadata
zohoDatatemplateID: “nocvr5515a”
zohoClientId: "your-client-id"
zohoClientSecret: "your-client-secret"
zohoFolderId: "67890"
zohoScope: "WorkDrive.files.CREATE,WorkDrive.DataTemplates.CREATE"

Output Example:

Result: "Upload complete. Success: 3, Failed: 1"
DetailedLog:
[OK] Document : Invoice_001.pdf
PDF Size: 1,024 bytes
PDF ID: 1234567890
JSON ID: 0987654321

[FAIL] Document : Invoice_002.pdf
Error : "KTA returned empty PDF content."

- Document upload to Zohoworkdrive along with extracted metadata fields from TTA



Security & Best Practices

- Input Validation: Always validate all input parameters before execution.
- Error Logging: Use DetailedLog for troubleshooting and audit trails.
- Token Management: Store Zoho credentials securely; never hard-code secrets.
- Testing: Test the integration in a non-production environment before go-live.
- Documentation: Maintain up-to-date documentation for all custom integrations.

Appendix: Private Methods & Classes

Private Methods:

- RunUploadAsync: Orchestrates the upload process.
- GetFolderDocuments: Retrieves documents from KTA.
- GetDocumentData: Fetches document metadata.
- GetDocumentAsPdf: Converts documents to PDF.
- GetZohoToken: Handles Zoho OAuth token acquisition.
- UploadToZoho: Uploads files to Zoho WorkDrive.
- BuildMetadataJson: Generates JSON metadata.

Private Classes:

- KtaConfig: Stores KTA configuration.
- ZohoConfig: Stores Zoho configuration.
- FolderDocument: Represents a KTA folder document.
- DocumentData: Represents document metadata.
- UploadItem: Represents upload results.

Summary

By following this guide, you can securely and efficiently deploy the ZohoUploader.dll in TotalAgility Cloud, enabling automated document uploads to Zoho WorkDrive with comprehensive logging and error free experience.